

Project Design Phase
Problem – Solution Fit

Date	19/06/25
Team ID	LTVIP2026TMIDS76471
Project Name	Exploratory Analysis of Rainfall Data in India for Agriculture
Maximum Marks	2 Marks

Problem – Solution :

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ☐ Understand the existing situation in order to improve it for your target group.

Template:



Define CS, TR into Focus on J&P, tag into BE, understand	1. CUSTOMER SEGMENT(S) CS - Utility company decision-makers - Government policymakers (energy departments) - Energy analysts and researchers - Public sector monitoring authorities	6. CUSTOMER CC - Limited technical/data visualization skills - Budget constraints for tool adoption - Reliance on manual Excel-based workflows - Limited access to cleaned, centralized data - Low IT infrastructure in smaller utility companies	5. AVAILABLE SOLUTIONS AS - Static government reports in PDF/Excel - Manual data analysis using spreadsheets - Internal dashboards with limited scope Pros: Familiar tools, simple setup Cons: No interactivity, slow, difficult to analyze, lacks filtering	Explore AS, Focus on AS, TR into BE, understand
	2. JOBS-TO-BE-DONE / PROBLEMS J&P - Understand state-wise and sector-wise electricity usage patterns - Forecast demand for better grid management - Identify peak hours and plan energy-saving programs - Analyze seasonal usage trends and post-lockdown impacts - Make data-driven decisions from raw usage data	9. PROBLEM ROOT CAUSE RC - No centralized platform for data-driven electricity consumption insights - Datasets are raw, unfiltered, and not visualized - Decision-makers lack tools and training to interpret the data easily - Growing complexity in managing supply-demand post-COVID and climate events	7. BEHAVIOUR BE - Use Excel to sort and manually analyze usage - Request reports from IT/data teams - Refer to government portals for downloads - Discuss patterns informally within departments - Use experience-based intuition over data evidence	
Identify strong TR & EM	3. TRIGGERS TR - External pressure from government mandates, public reports, or new datasets requiring improved energy planning and transparency. - Operational challenges like blackouts, peak season budgeting, or rising interest in sustainability prompt action from utility stakeholders.	10. YOUR SOLUTION SL A web-based dashboard using Tableau embedded into a Task app. The processed data stored in MySQL, integrated with real-time filtering. Visualizations include: Time-wise, region-wise, lockdown comparison, and top-10 urban usage charts. Interactive filters for users to select year, region, and time period. Optional ML-powered demand forecasting. Published on Tableau Public for easy access and sharing.	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE Download datasets from energy portals (POKCO, Ministry of Power) Read insights or trends from news portals or LinkedIn Watch dashboard demos (YouTube, Tableau Public) 8.2 OFFLINE Attend government meetings Internal review meetings and printed reports Collaborate on planning documents manually	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER EM Before: Overwhelmed, frustrated, uncertain, data-blind After: Informed, empowered, confident, able to make smart decisions			

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	17/06/25
Team ID	LTVIP2026TMIDS76471
Project Name	Exploratory Analysis of Rainfall Data in India for Agriculture
Maximum Marks	4Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Import	Import data from CSV Live database integration (MySQL)
FR-2	Data Cleaning & Transformation	Clean missing values Add calculated fields like Year, Lockdown
FR-3	Data Visualization	Create Tableau worksheets Build multiple dashboards
FR-4	User Interaction	Filter by region, year View comparative bar charts Analyze pre/post lockdown trends
FR-5	User Access	Role-based view (Analyst, Policy Maker, Developer) Download/export insights
FR-6	Feedback Loop	Stakeholders review and request changes Implement revision cycle

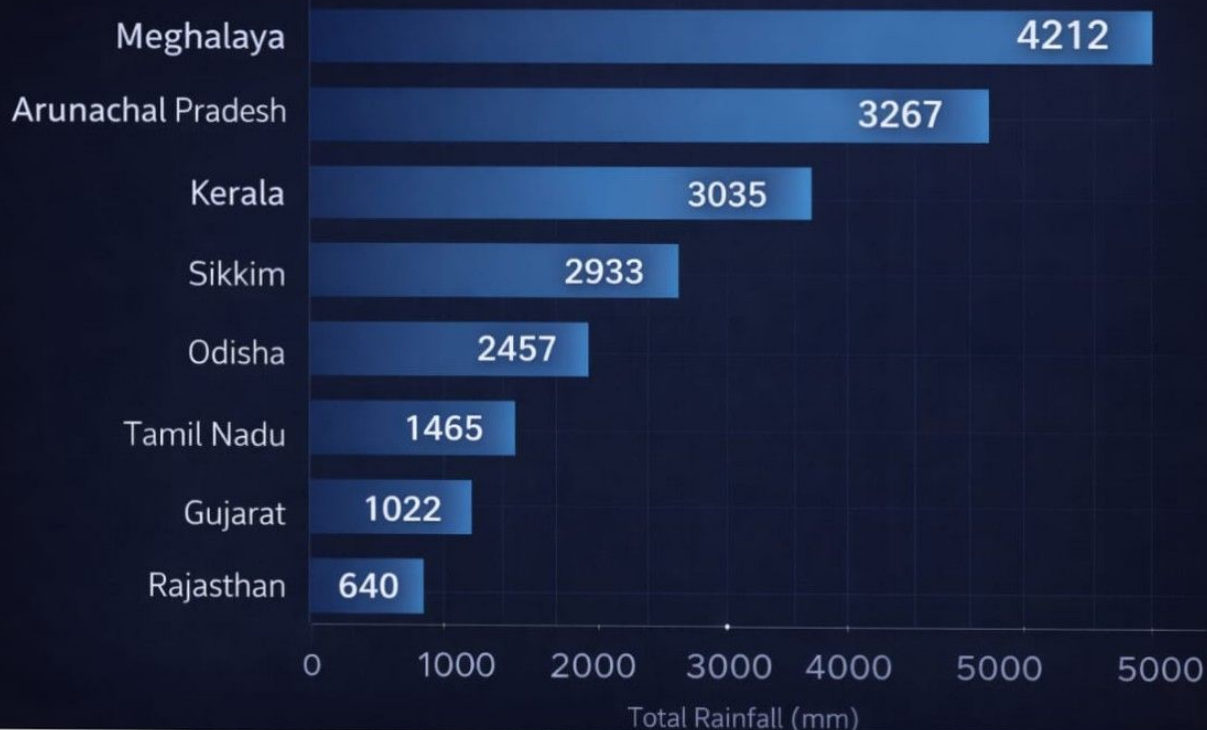
Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Intuitive dashboard with clear filters and legends
NFR-2	Security	Role-based access and secure database connection
NFR-3	Reliability	System handles unexpected data formats and ensures accuracy
NFR-4	Performance	Fast load times and responsiveness of visualizations
NFR-5	Availability	Dashboard is accessible across devices with minimal downtime
NFR-6	Scalability	Supports increasing data volume and new features

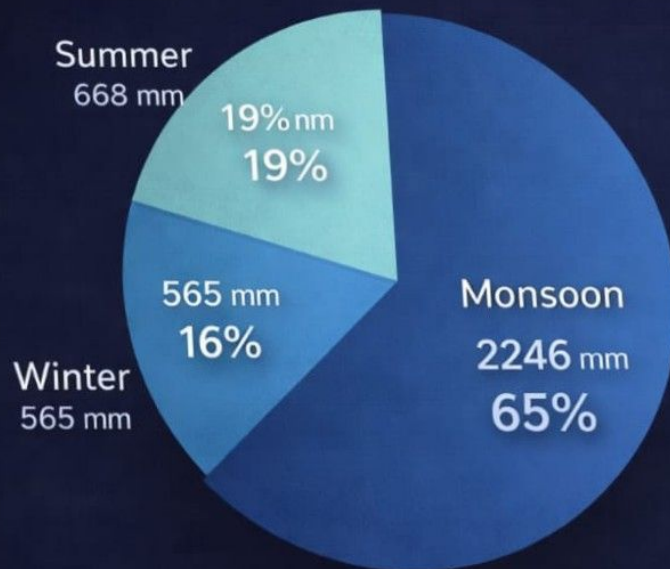
Bar Chart:

State-wise Annual Rainfall

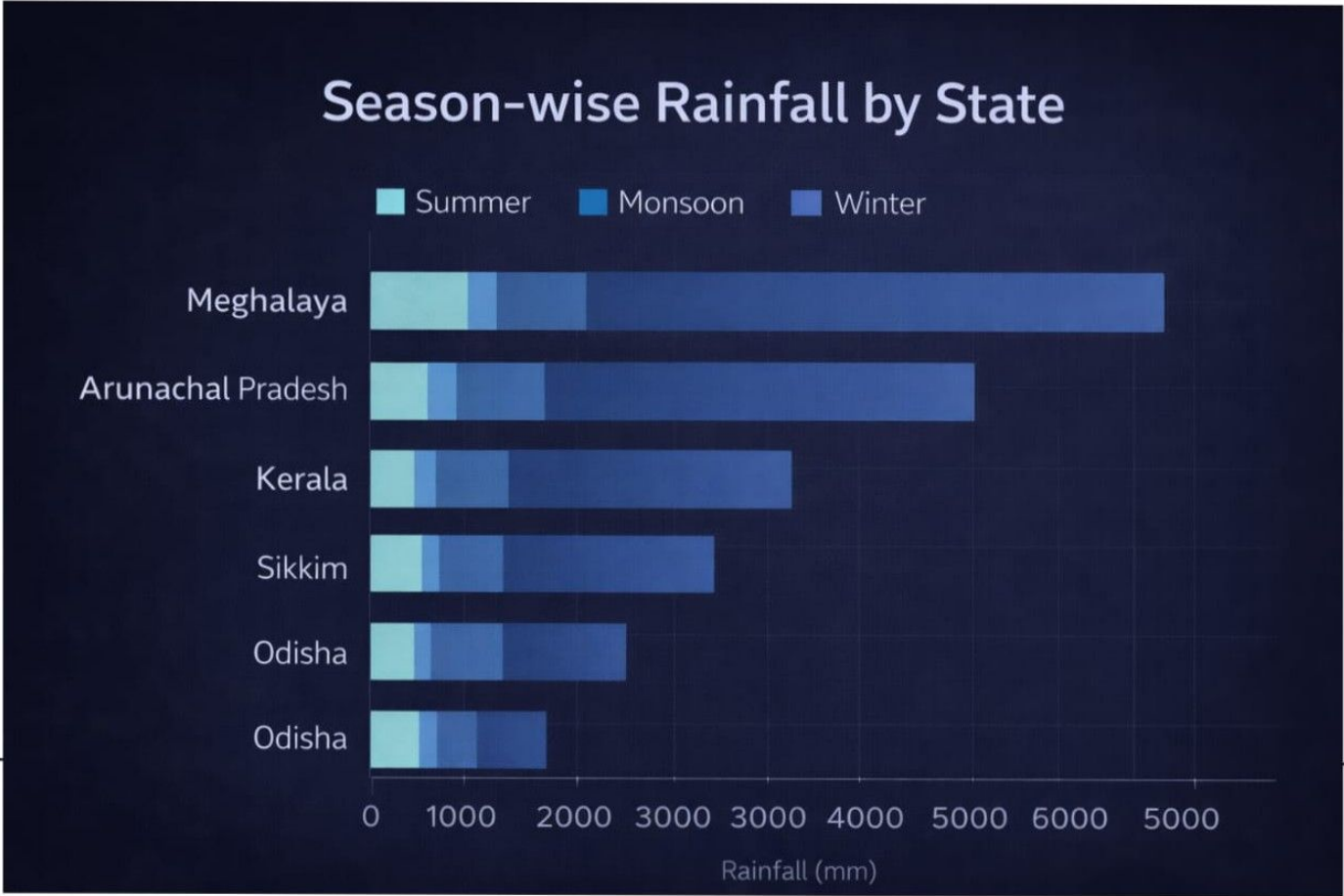


Pie Chart:

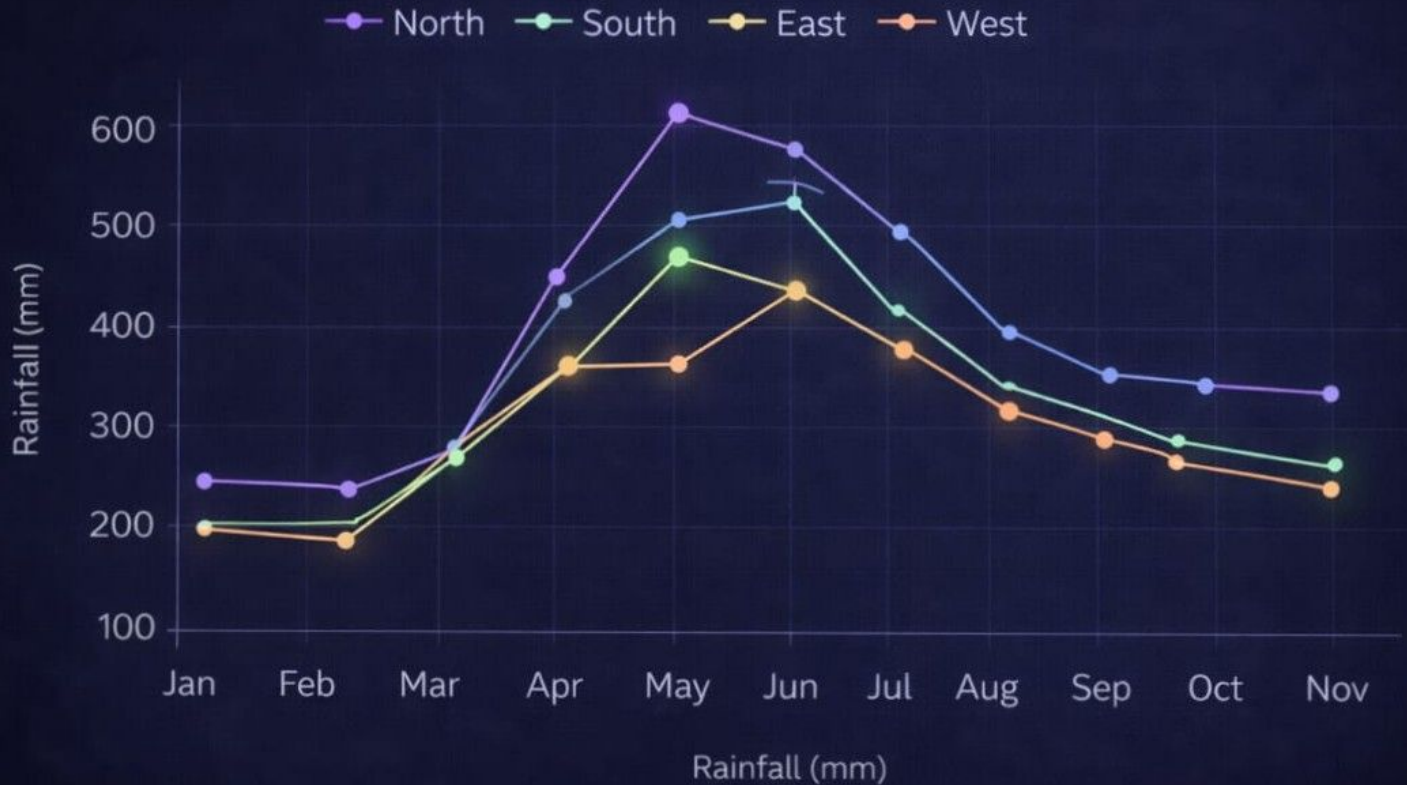
Season-wise Rainfall Distribution



Stacked Bar Chart:



Monthly Rainfall Trend in 2023



Bubble Chart:

